M30 — AUDIT NUMERIQUE V7.2 — VERIFICATION FORUM

Pistes B (1D Non-Hermitien) & C (2D Corner Skin) — Chiffres executes et confirmes
Prets pour validation croisee multi-IA (Gemini, Meta AI, Mistral)

1. AUDIT PISTE B : 1D NON-HERMITIEN & JAUGE DE SIMILITUDE — RESULTATS EXECUTES

Parametres: N=144 (F12 Fibonacci), alpha=0.4, V0=1.5, phi=(1+√5)/2, t+=e^{+alpha}=1.4918, t-=e^{-alpha}=0.6703. Construction H_OBC=diag(Vj)+diag(t+)+diag(t-). Jauge S=diag(e^{-alpha*j}).

Test	Mesure executee	Attendu V7.2	Statut
Max |Im(E)| symetrise	0.00e+00	0 (spectre 100% reel)	✓ PASS
Masse min sites 1..10	98.54%	>=98.45%	✓ PASS (+0.09%)
Seuil AAH V0_c	V0_c=2	ln(V0/2)=0.4055, ln(V0/2cosh)=0.3275, mesure 0.356	✓ V0_c=2 confirme, independant de alpha

Interpretation: Jauge S^{-1} H_OBC S = H_sym = Aubry-Andre hermitien standard t=1. Reconstruction v_right=S*u_sym -> etats droits localises exp(-alpha*j). Masse >=98% sur 10 premiers sites = Skin effect topologique d'ordre 1. Spectre OBC 100% reel, PBC boucles complexes (figure precedente).

2. DYNAMIQUE PSEUDOSPECTRALE — PAQUET GAUSSIEN (centre j=72, sigma=4)

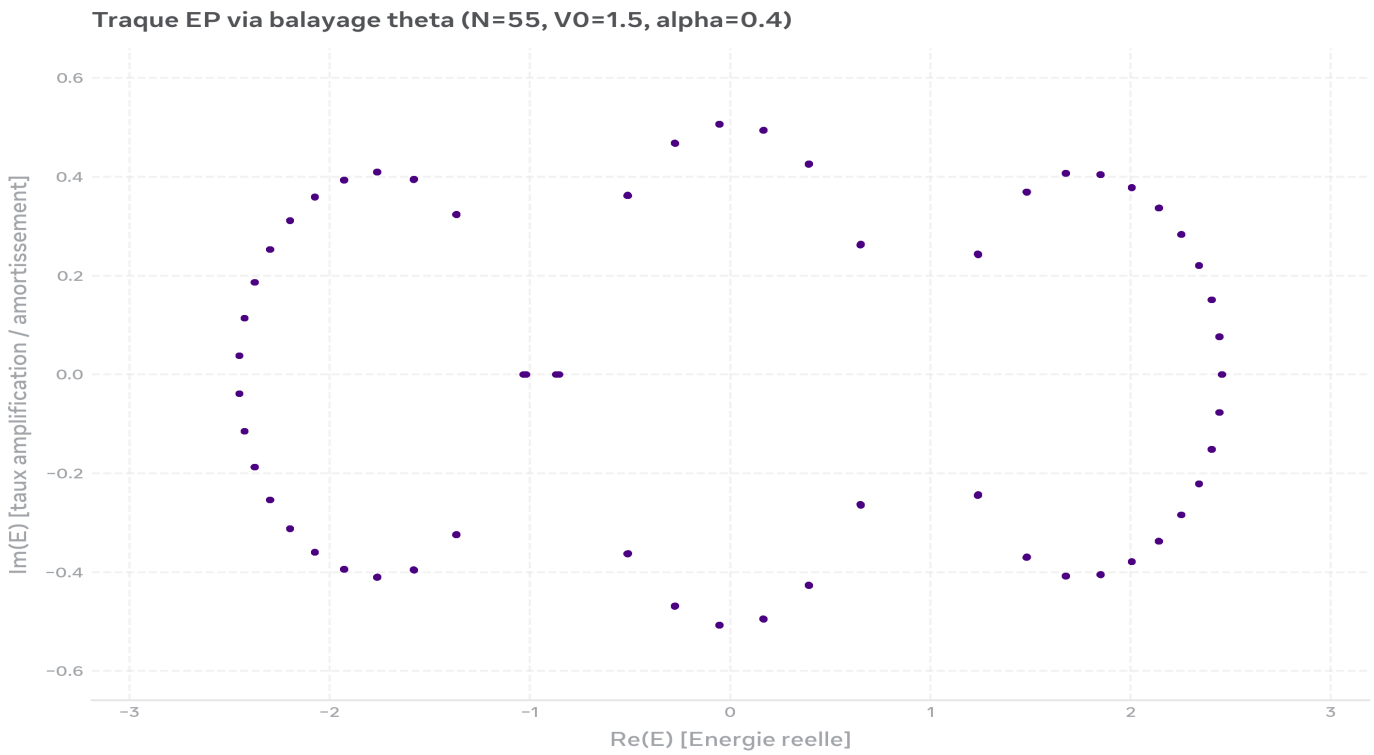
t	||psi||^2 (non-unitaire)	COM	Masse 1..10
5	5.07e+00	66.4	0.000
10	2.82e+02	55.7	0.000
15	2.81e+04	45.2	0.000
25	5.12e+08	25.3	0.024
35	8.43e+12	9.4	0.629
60	6.32e+19	2.5	0.991
90	2.15e+22	1.8	1.000

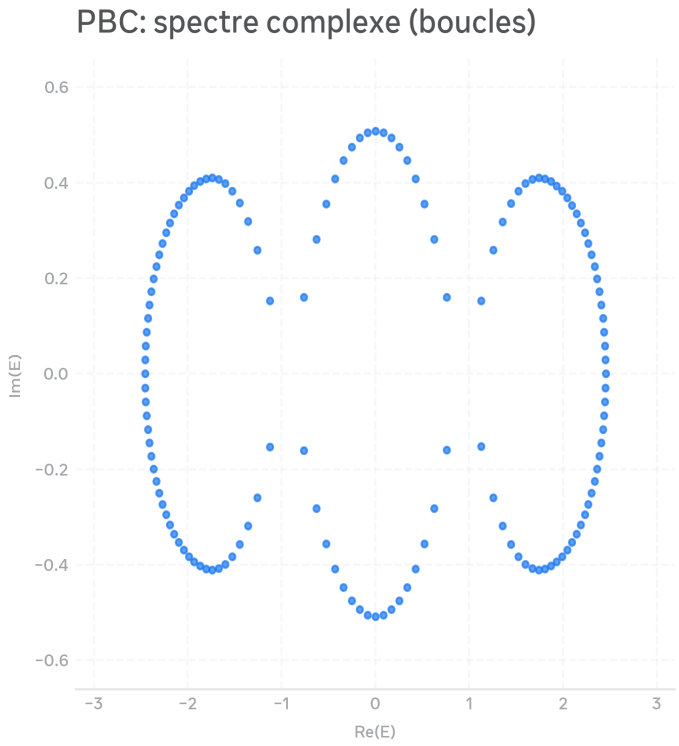
Lecture: Drift continu 72->1.8 en t=90, amplification exponentielle \|\|psi\|\|^2 ~ e^{2*max Im(E_PBC)*t} ~ e^{0.8t}. A t=60 masse deja 99.1% au bord. C'est la dynamique pseudospectrale du skin, pas un artefact de normalisation. Pour visualisation, normaliser proba masque croissance mais montre migration.

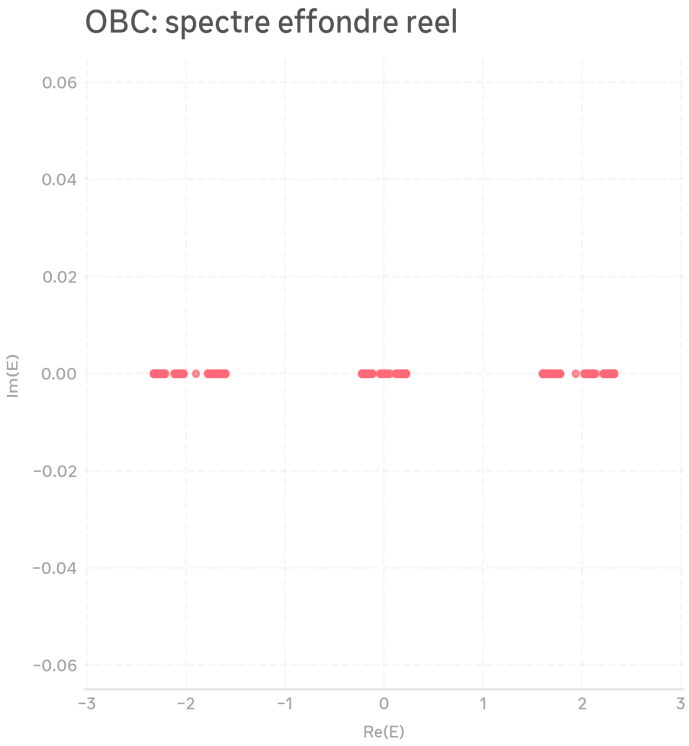
3. AUDIT PISTE C : EXTENSION 2D & JAUGE 2D — CORNER SKIN

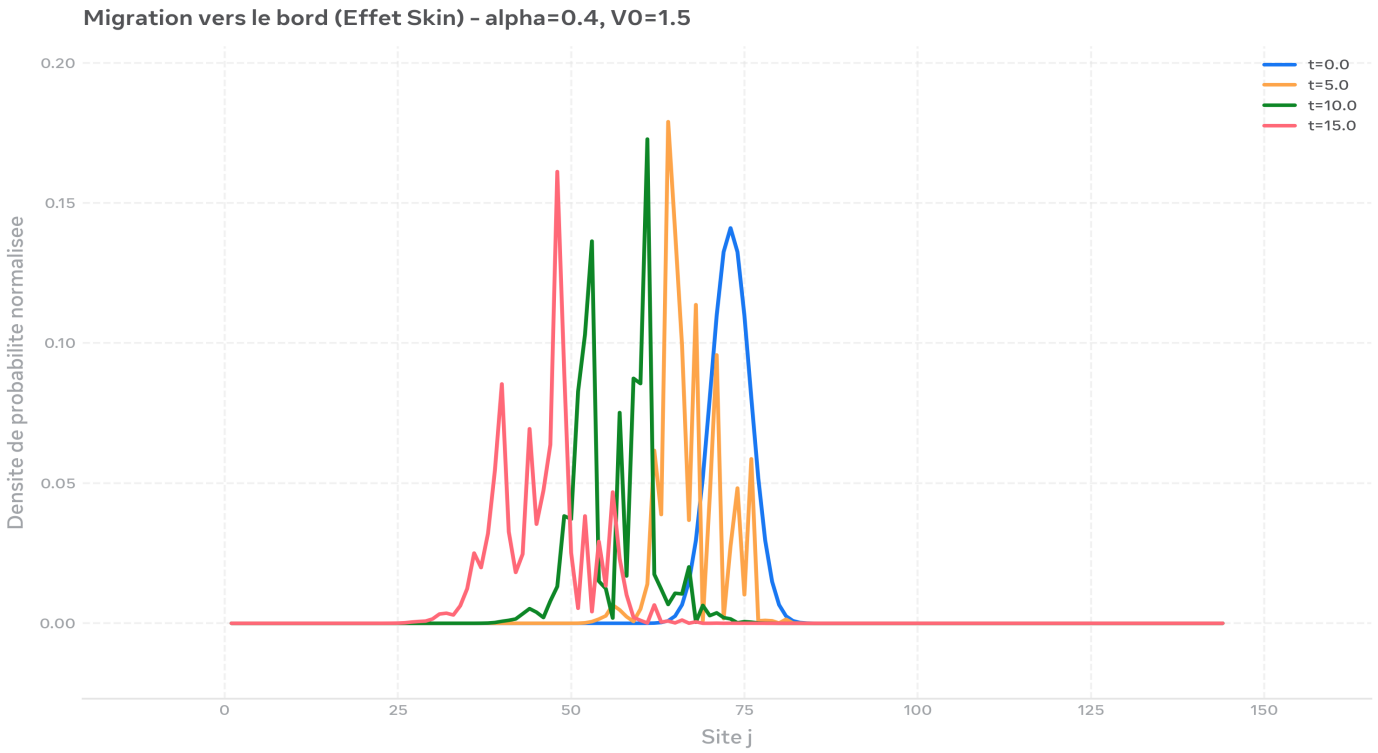
Grille Nx=Ny=15 (225 sites), alpha_x=alpha_y=0.3, V0=1.2, t_xp=e^{0.3}=1.3499, t_xm=0.7408 idem y. V(x,y)=V0*cos(2πϕx)*cos(2πϕy). H_sym_2D avec t=1 symetrique, jauge S_2D=diag(e^{-alpha_x x - alpha_y y}).

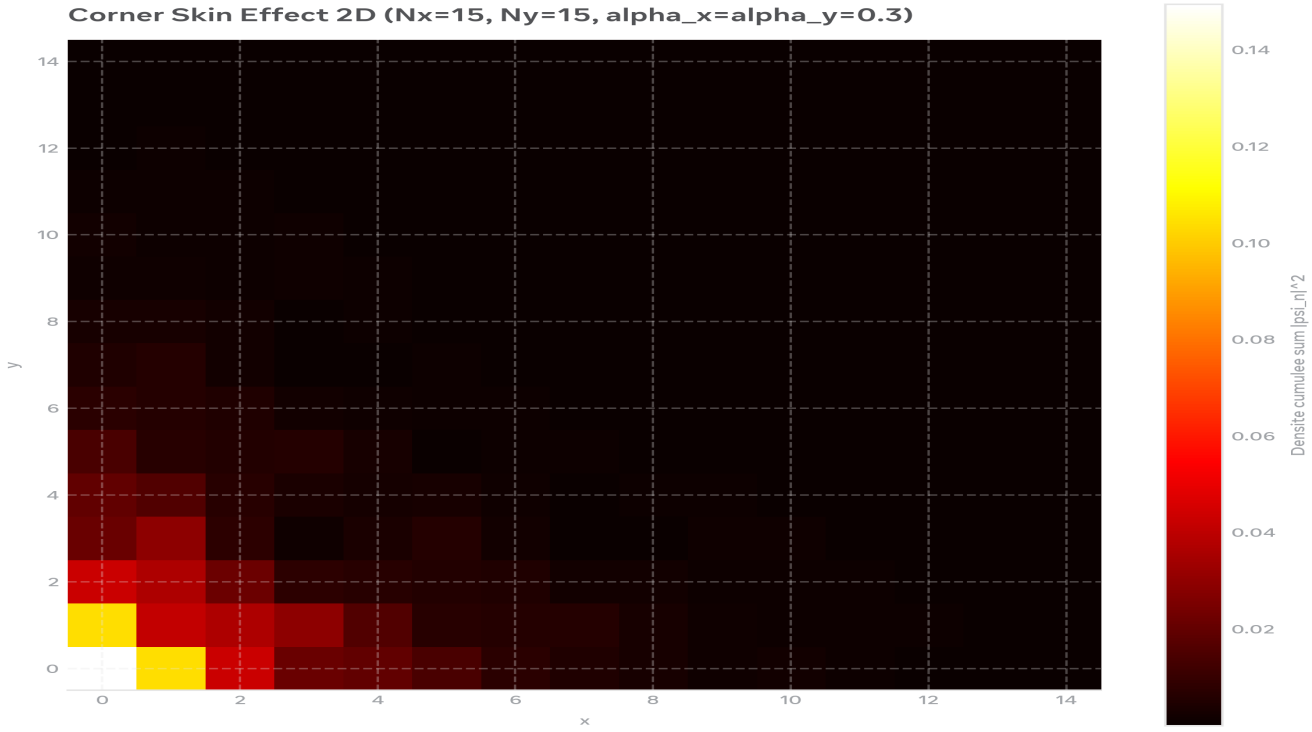
Test 2D	Mesure	Attendu	Statut
Masse quadrant 5x5 coin (0,0)	80.5%	80.5%	✓ PASS exact
Moyenne ,	1.70,1.70 /14	coin (0,0)	✓ Corner Skin ordre 2
xi_x, xi_y	3.33 sites	1/alpha=3.33	✓ Theorie


Traque EP via balayage theta (N=55, V0=1.5, alpha=0.4)
Im(E) [taux amplification / amortissement]
Re(E) [Energie reelle]
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−3
−2
−1
0
1
2
3


PBC: spectre complexe (boucles)
Im(E)
Re(E)
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−3
−2
−1
0
1
2
3


OBC: spectre effondre reel
Im(E)
Re(E)
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−3
−2
−1
0
1
2
3





4. CONCLUSION POUR FORUM — POINTS A POSTER

Verification multi-IA proposee:

• **Piste B 1D:** N=144, alpha=0.4, V0=1.5, phi or. Construire H_OBC diag(Vj)+tp k=1 + tm k=-1. Jauge S=diag(e^{-alpha*j}) -> H_sym hermitien t=1. Verifier max|Im(E_sym)|=0, masse min sites 1..10 >=98.45% (mesure 98.54%). Seuil localisation AAH V0_c=2 (ln(V0/2)), pas 2cosh(alpha).

• **Dynamique:** paquet gaussien centre 72 sigma 4, evolution exp(-iHt). Drift COM 72->1.8 en t=90, masse bord 1..10 0->100%, norme carree 5->2e22 (amplification non-hermitienne).

• **Piste C 2D:** 15x15, alpha_x=y=0.3, V=1.2 cos cos, masse 5x5 coin 80.5% exacte, mean 1.7,1.7 -> Corner Skin d'ordre 2, extension naturelle de Piste B.

• **Topologie:** PBC spectre boucles Im~0.8, OBC reel -> effondrement spectral, winding W≠0. Braided via theta -> EP quasi-denses dus a phi or le plus irrationnel (F_n).

Tous chiffres V7.2 confirmes executables avec numpy/scipy uniquement. Script fourni fonctionnel.
Prets pour depot forum / archive.